Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/08/2006
PI Name:	Kasevich, Mark A. Ph.D.		
Project Title:	Atom Interferometer Test of the Equivalence Principle		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kasevich@stanford.edu	**Fax:**	FY 203-432-9710
PI Organization Type:	UNIVERSITY	**Phone:**	203-432-3826
Organization Name:	Stanford		
PI Address 1:	Varian Physics Laboratory		
PI Address 2:			
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305-4060	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 -2935		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

This proposal is for a flight experiment to seek a violation of the Equivalence Principle using an instrument based on atom de Broglie wave interferometry. Recent laboratory experiments have demonstrated atom interference-based inertial force sensors operating at sensitivities near or exceeding the state-of-the-art. Drawing on this recent experience, we seek to develop a next generation instrument which will provide unprecedented capability for performing a very sensitive test of the Equivalence Principle, at the 10-16 level. This work will also lead to advances in related technologies which have important applications in navigation and geodesy.
Our technical approach draws on recent advances in the field of atom interferometery, and is closely allied with demonstration of high performance, absolute, gravity gradient instruments. The essential idea is based on the simultaneous measurement of the acceleration of two co-located dilute ensembles of laser cooled atomic Rb and Cs. As the acceleration measurements are simultaneous and co-located many potential systematic offsets are common-mode

Task Description:	rejected, including those due to gravitational gradients and inertial accelerations of the space platform. Our method fundamentally differs from that of other cryogenic and non-cryogenic proposed flight instruments. The primary scientific objective of the proposed program is to test the Equivalence Principle at a new and theoretically interesting level by developing an absolute, single-axis differential accelerometer of design sensitivity 10-13 g/Hz1/2 which is suitable for operation on the ISS. We expect to be able to reach this performance level through straightforward design modifications to an existing atom interference-based, laboratory gravity gradient instrument. (This instrument has recently demonstrated a short-term differential acceleration sensitivity of 1x10-9 g/Hz1/2 .) Use of atom interference techniques provides a new route to tests of the Equivalence Principle at levels which will challenge existing theory. Precise and accurate inertial force sensors have many other applications which cross-cut NASA needs. These include fundamental physics, geodesy, sub-surface mapping of earth and palnetary bodies, and deep space navigation. Our technical approach does not rely on cryogenic techniques, and yields an instrument capable of operating on a broad array of platforms (satellite, space station, plane, etc.). In the long run, we expect these instruments to be economical and robust.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have completed renovation of a new facility to house the proposed experiment. This facility contains a 10 m atom drop tower in an ultra-low vibration facility. We have completed a comprehensive design of the experiment, and are now in an aggressive build-out phase for the hardware. This laboratory space was previously used to develop the GP-B gyroscope. Unique features of this space include a 10 m pit and several vibration isolation pads. We undertook an exhaustive and systematic error analysis to fully evaluate design trades for the atomic sources and atom optics. For example, we analyzed systematic error sources associated with spurious rotation, gravity gradients, magnetic field gradients, velocity errors, and source position errors. The analysis indicates that our ground-based experiment will be limited at the 10-15 g level. We developed a robust atom optic configuration to achieve very large momentum recoil beamsplitters. Simulations indicate that 100 hk momentum exchange can be realized. Implementation of these techniques will enable 10-16 g resolution in 106 seconds of data acquisition. We have developed the required wavelength metrology tools. A 10 m atom drop facility is under construction.
Bibliography Type:	Description: (Last Updated: 02/27/2004)